Characterization of Mach-Zehnder Interferometer Based on SiEPIC PDK

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***Abstract*— This paper provides a brief overview of the design of a Mach-Zehnder Interferometer. Experimental results show the relationship between the free spectral range of the device and the difference in path length in TE mode.**

***Index Terms*—Mach-Zehnder Interferometer, Silicon Photonics**

I. INTRODUCTION

THE Mach–Zehnder Interferometer (MZI) is a widely used optical circuit with numerous applications, including optical switching and modulation. In this paper, we propose a Mach–Zehnder interferometer design and characterize its performance by introducing a controlled mismatch in the waveguide arm lengths.

II. THEORY

A. Waveguide

We consider a lossless waveguide with length L, effective refractive index n_{eff}, and the propagating wave has a wavelength λ and electric field E. The propagation constant is defined as

$$\beta = \frac{2\pi n_{eff}}{\lambda} \quad (1)$$

The propagation of plane wave through the waveguide is then defined as

$$E_o = Ee^{-i\beta L} \quad (2)$$

B. Y-Branch Splitter/Combiner

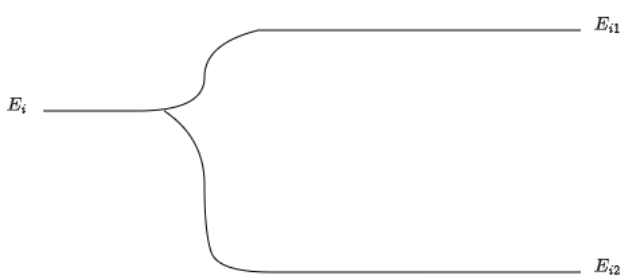

Fig. 1. Y-Branch Splitter

We consider an ideal Y-Branch splitter, as shown in **Fig. 1**. For an input intensity of I_i and electric field E_i, the light splits equally between the two branches, resulting in the output intensity

$$I_{i1} = I_{i2} = \frac{I_i}{2} \quad (3)$$

and electric field

$$E_{i1} = E_{i2} = \frac{E_i}{\sqrt{2}} \quad (4)$$

Similarly, for an ideal Y-branch combiner, the resulting electric field output from two inputs will be

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} \quad (5)$$

and the output intensity will depend on the phase difference of the two inputs. This will be discussed in the Mach-Zehnder Interferometer section.

C. Mach-Zehnder Interferometer (MZI)

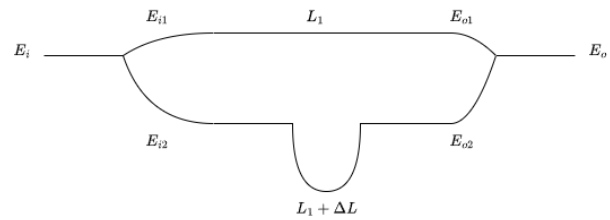

Fig. 2. Proposed MZI design

The proposed MZI design shown by **Fig. 2** utilizes one Y-branch splitter, one Y-branch combiner, and two waveguides. A path difference is introduced to the two waveguides for the purpose of generating a phase difference and assist the characterization of the free spectral range of the MZI. Starting from the input, using equation (4), we get

$$E_{i1} = E_{i2} = \frac{E_i}{\sqrt{2}} \quad (6)$$

Substitute into equation (2),

$$E_{o1} = E_{i1}e^{-i\beta_1 L_1} = \frac{E_i}{\sqrt{2}}\, e^{-i\beta_1 L_1} \quad (7)$$

$$E_{o2} = E_{i2}e^{-i\beta_2 (L_1+\Delta L)} = \frac{E_i}{\sqrt{2}}\, e^{-i\beta_2 (L_1+\Delta L)} \quad (8)$$

Finally, from equation (5),

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}\left(e^{-i\beta_1 L_1} + e^{-i\beta_2 (L_1+\Delta L)}\right) \quad (9)$$

Thus, the intensity of the output is

$$I_o = \frac{I_i}{4}\left|e^{-i\beta_1 L_1} + e^{-i\beta_2 (L_1+\Delta L)}\right|^2$$

$$= \frac{I_i}{2}[1 + \cos(\beta_1 L_1 - \beta_2(L_1 + \Delta L)](10)$$

We can assume that the two wave guides have the same properties, hence the same propagation constant. Thus, the output intensity can be rewritten as

$$I_o = \frac{I_i}{2}[1 + \cos(\beta \Delta L)] = \frac{I_i}{2}[1 + \cos(\delta)] \ (11)$$

where $\delta = \beta \Delta L$.

D. Free Spectral Range(FSR)

The FSR represents the period of oscillation of the MZI transfer function with respect to wavelength. It can be approximated as

$$FSR = \frac{\lambda^2}{\Delta L n_g} \ (12)$$

where n_g is the group index of the waveguide. It is related to the velocity of the beam travelling inside the waveguide depending on the wavelength of the beam.

III. MODELLING AND SIMULATION

The modeling and simulations in this work are based on the SiEPIC PDK. The modeling and simulation are performed using Lumerical MODE and Lumerical INTERCONNECT. The layout is done by KLayout. The design will be optimized for a $1550nm$ light source to operate in single mode. The bending radius is set to $500nm$ for all designs.

A. Waveguide

The design is a Silicon on Insulator (SOI) slab waveguide. This paper goes for a strip waveguide for ease of fabrication. The rectangular waveguide has a standard $220nm$ thickness and $500nm$ width.

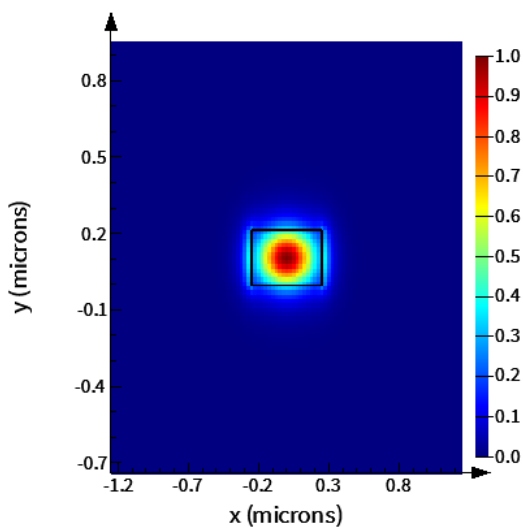

Fig. 3. TE mode electric field intensity of designed waveguide

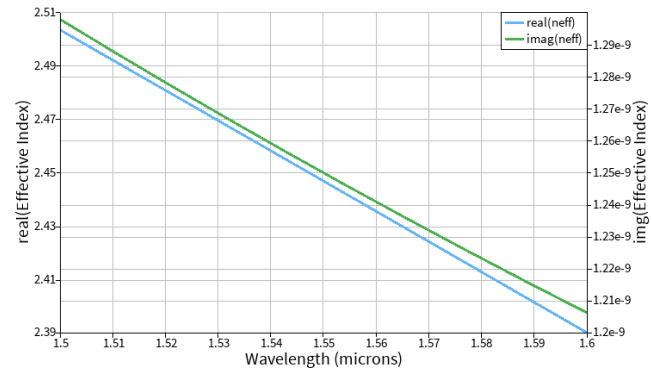

Fig. 4. Effective index relative to wavelength

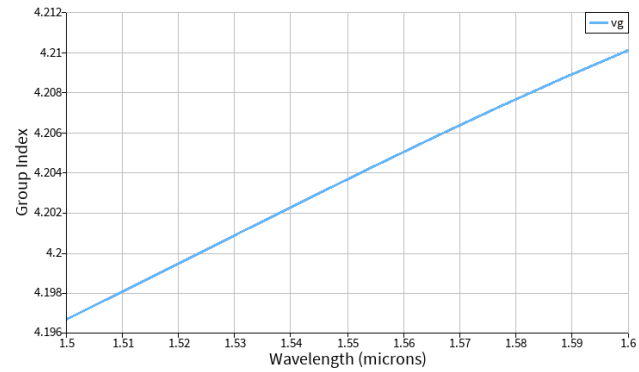

Fig. 5. Group index relative to wavelength

The waveguide compact model is derived from Taylor expansion around the center wavelength. The fitted model found to be

$$n_{eff} = 2.45 - 1.13(\lambda - 1.55) - 0.04\,(\lambda - 1.55)^2 \ (13)$$

B. Mach-Zehnder Interferometer

The design is implemented and simulated in Lumerical INTERCONNECT. Fiber grating couplers were added.

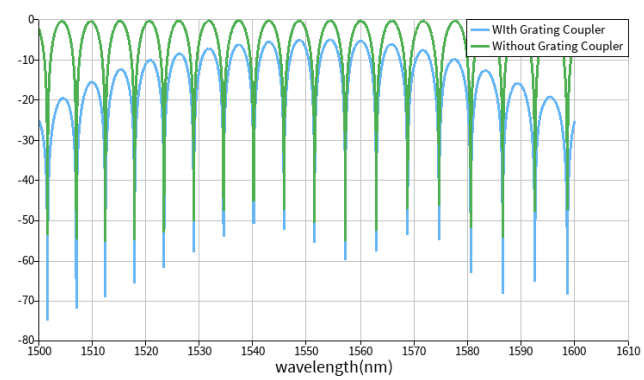

Fig. 6. Spectrum of MZI with $\Delta L = 100nm$

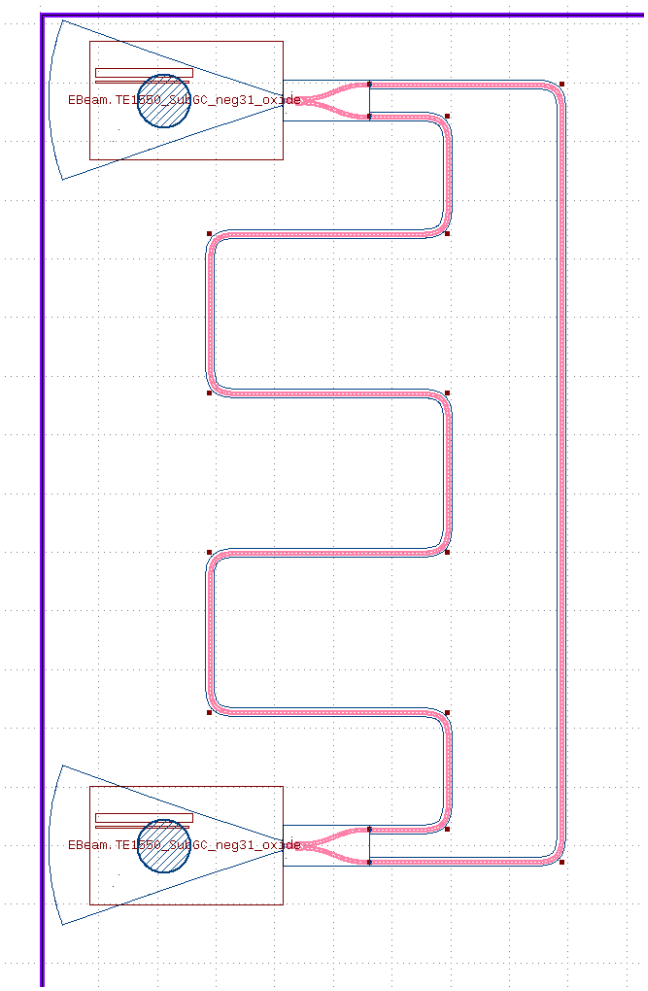

Fig. 7. Proposed imbalanced MZI layout

Same architecture is used for all MZIs to offer a better comparison. Two separate designs using directional and adiabatic couplers are also tested with $\Delta L = 100\mu m$.

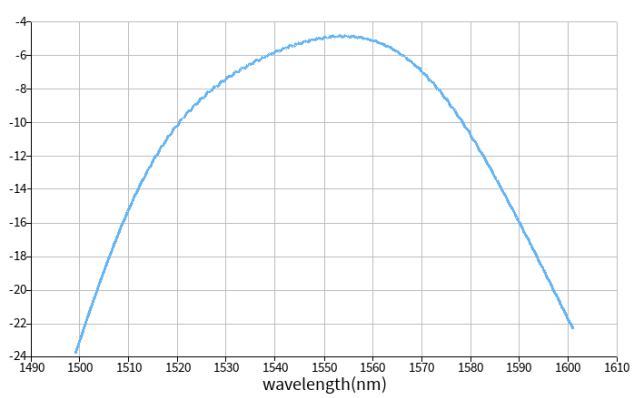

Fig. 8. Simulated insertion loss of fibre grating coupler

The insertion loss of the fibre grating coupler is $4.91dB$.

TABLE I
SIMULATED THEORETICAL FSR

$\Delta L(\mu m)$	$FSR(nm)$
100	5.717
200	2.894
300	1.895
400	1.431
500	1.128

TABLE II
SIMULATED FSR FOR LAYOUT

$\Delta L(\mu m)$	$FSR(nm)$
100	5.607
200	2.900
300	1.919
400	1.449
500	1.143

TABLE III
SIMULATED THEORETICAL FSR FOR COUPLED OUTPUT, $\Delta L = 100\mu m$

Coupler Type	$FSR(nm)$
Directional	5.820
Adiabatic	6.389

TABLE IV
SIMULATED LAYOUT FSR FOR COUPLED OUTPUT, $\Delta L = 100\mu m$

Coupler Type	$FSR(nm)$
Directional	5.800
Adiabatic	5.648

Comparing the simulated results from Table I and Table II, we conclude that the layout demonstrates the expected behavior of the designed imbalanced MZI.

IV. Conclusion

Tentative

V. Acknowledgment

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Henry Wu (Member, IEEE), photograph and biography not available at the time of publication.